Course code	**Course Title**	**L**	**T**	**P**	**C**
BCSE332L	**Deep Learning**	**3**	**0**	**0**	**3**
Pre-requisite	**NIL**	**Syllabus version**			
		1.0			

Course Objectives

1. Introduce major deep neural network frameworks and issues in basic neural networks.
2. To solve real world applications using Deep learning.

Course Outcomes

At the end of this course, student will be able to:

1. Understand the methods and terminologies involved in deep neural network, differentiate the learning methods used in Deep-nets.
2. Identify and apply suitable deep learning approaches for given application.
3. Design and develop custom Deep-nets for human intuitive applications.
4. Design of test procedures to assess the efficiency of the developed model.
5. To understand the need for Reinforcement learning in real – time problems.

Module:1	**Introduction to neural networks and deep neural networks**	**7 hours**
Neural Networks Basics - Functions in Neural networks – Activation function, Loss function - Function approximation - Classification and Clustering problems - Deep networks basics - Shallow neural networks – Activation Functions – Gradient Descent – Back Propagation – Deep Neural Networks – Forward and Back Propagation – Parameters – Hyperparameters.		
Module:2	**Improving deep neural networks**	**8 hours**
Mini-batch Gradient Descent – Exponential Weighted Averages – Gradient Descent with Momentum – RMSProp and Adam Optimization – Hyperparameter tuning – Batch Normalization – Softmax Regression – Softmax classifier – Deep Learning Frameworks – Data Augmentation - Under-fitting Vs Over-fitting.		
Module:3	**Convolution neural networks**	**6 hours**
Foundations of Convolutional Neural Networks – CNN operations – Architecture – Simple Convolution Network – Deep Convolutional Models – ResNet, AlexNet, InceptionNet and others.		
Module:4	**Recurrent networks**	**6 hours**
Recurrent Neural Networks - Bidirectional RNNs, Encoder, Decoder, Sequence-to-Sequence Architectures, Deep Recurrent Networks, Auto encoders - Bidirectional Encoder Representations from Transformers (BERT).		
Module:5	**Recursive neural networks**	**6 hours**
Long-Term Dependencies - Echo State Networks - Long Short-Term Memory and Other Gated RNNs - Optimization for Long-Term Dependencies - Explicit Memory.		
Module:6	**Advanced Neural networks**	**6 hours**
Transfer Learning – Transfer Learning Models – Generative Adversarial Network and their variants – Region based CNN – Fast RCNN - You Only Look Once – Single shot detector.		
Module:7	**Deep reinforcement learning**	**5 hours**
Deep Reinforcement Learning – Q-Learning – Deep Q-Learning – Policy Gradients - Advantage Actor Critic (A2C) and Asynchronous Advantage Actor Critic (A3C) – Model based Reinforcement Learning – Challenges.		
Module:8	**Contemporary issues**	**1 hour**
	Total Lecture hours:	**45 Hours**

Text Book(s)